

TIC 237226943

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.998340 d	VSX	no match
Epoch (t0)	1765.9972 BTJD	ASAS-SN	no match
Duration	0.0840 d	Gaia eclipsing-binary table	no match
Depth	1.018 % (10,182 ppm)	Gaia variable classification	no match
SNR	168.5	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	318 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.35 (gate >= 0.50)		
Robust depth sigma	161.1 (gate >= 7.0)		
Odd/even z-score	1.94 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

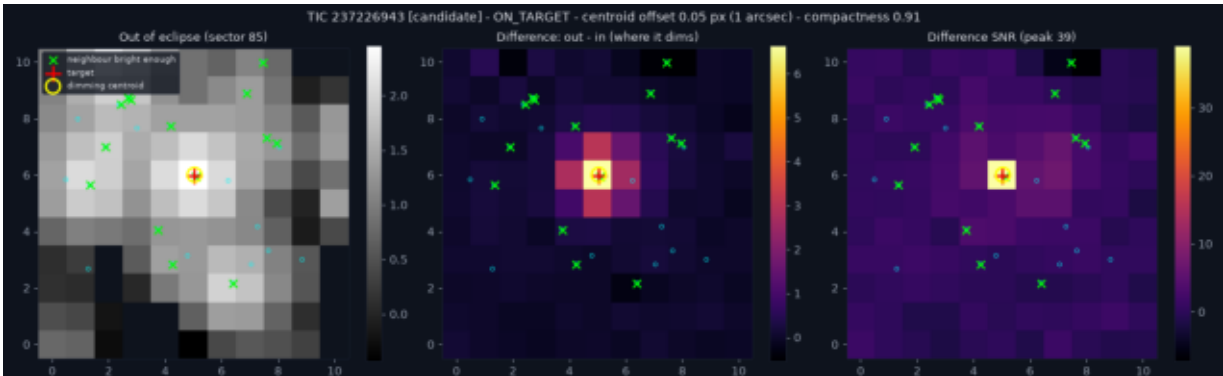
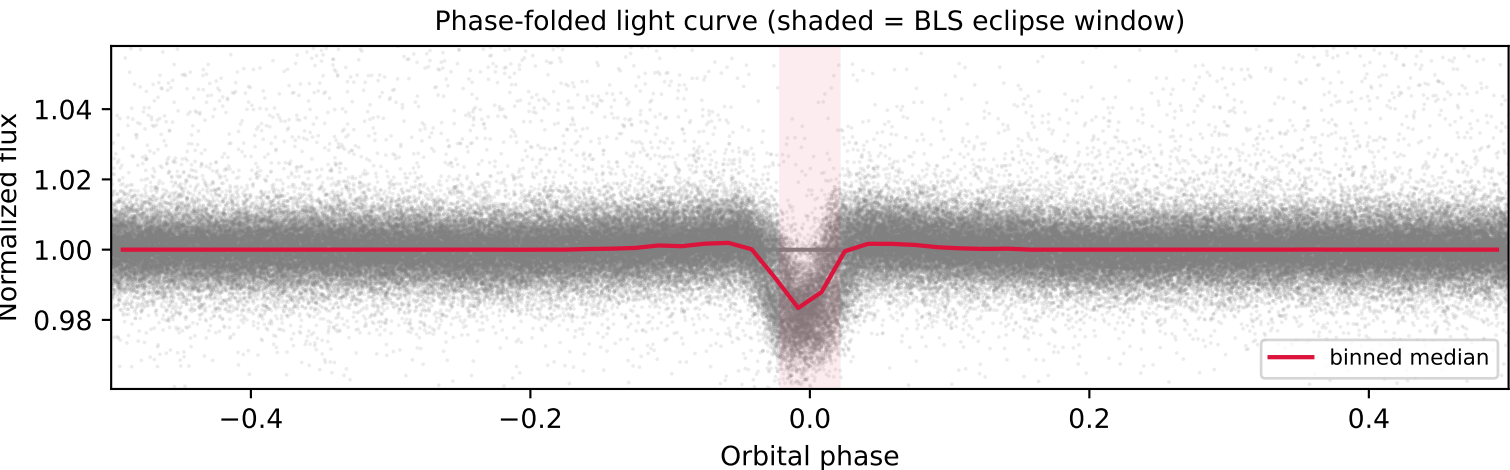
Sector 85 light-curve dip in this sector: 44.0 sigma t0 refined by -0.0150 phase

Difference-image peak SNR 39.20 (gate >= 5.0) Compactness 0.91 (gate >= 0.50)

Dimming centroid offset 0.05 px (1 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1884574048

0.10 px away, dTmag +4.35 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.051 px (dimming centroid 0.049 px from target vs 0.101 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.